

Module 5: Lesson 3

Deep and Bidirectional LSTMs

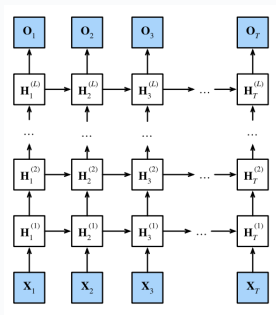


Outline

- ▶ Deep RNNs
- ▶ Bidirectional RNNs and Hidden Markov models
- ▶ Bidirectional LSTMs

Deep RNNs

Although we have practically already worked with RNNs with more than 1 hidden layer, we have not discussed them theoretically. This is indeed quite intuitive to see:



Where each hidden layer l will be expressed as:

$$H_t^{(l)} = \phi_l \left(H_t^{(l-1)} W_{xh}^{(l)} + H_{t-1}^{(l)} W_{hh}^{(l)} + b_h^{(l)} \right)$$

Bidirectional RNNs and Markov models

So far we have assumed that our goal is to model the prediction of the next observation of a sequence given the previous observations. But this can also work both-ways. A clear example arises when dealing with language models:

- ▶ I need to ___
- ▶ I need to ___ something, I'm starving!

What you be more likely to fill in the blank in the first or the second example? Obviously, when we have more information at our disposal, we tend to do better predictions. So does a RNN model!

In Section 10.4.1 of the book you will have an illustrative example of the implications of bidirectionality in Hidden Markov models. If forward (π_{t+1}) and backward (ρ_{t-1}) recursion are given by:

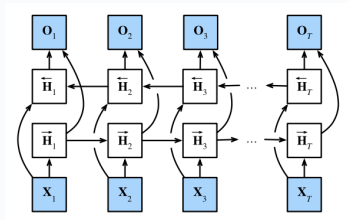
$$\pi_{t+1}(h_{t+1}) = \sum_{h_t} \pi_t(h_t) P(x_t|h_t) P(h_{t+1}|h_t) \qquad \rho_{t-1}(h_{t-1}) = \sum_{h_t} P(h_t|h_{t-1}) P(x_t|h_t) \rho_t(h_t)$$

Then, the conditional probability of a given observation j in the sequence is proportional to:

$$P(x_j|x_{-j}) \propto \sum_{h_j} \pi_j(h_j) \rho_j(h_j) P(x_j|h_j)$$

Bidirectional RNNs

How does bidirectionality work in the context of RNNs?



As you can see, it is relatively simple and intuitive. Similar to the Hidden Markov models, we will need to define the forward ($\vec{H}_t$) and backward ($\overleftarrow{H}_t$) hidden states, which we will do next.

Bidirectional RNNs were first introduced in a paper by Schuster and Paliwal (1997):

Schuster, Mike, and Kuldip K. Paliwal. "Bidirectional recurrent neural networks." *IEEE transactions on Signal Processing* 45.11 (1997): 2673-2681. [Paper here!](#)

Bidirectional RNNs

The forward and backward hidden states in a, say, bidirectional LSTM will be updated as follows:

$$\vec{H}_t = \phi \left(X_t W_{xh}^{(f)} + \vec{H}_{t-1} W_{hh}^{(f)} + b_h^{(f)} \right)$$

$$\overleftarrow{H}_t = \phi \left(X_t W_{xh}^{(b)} + \overleftarrow{H}_{t+1} W_{hh}^{(b)} + b_h^{(b)} \right)$$

This means that we will need to concatenate $\vec{H}_t$ and $\overleftarrow{H}_t$, so that the resulting hidden state, H_t would be of size $n \times 2h$. Once we have H_t , the output layer would be computed as usual:

$$O_t = H_t W_{hq} + b_q$$

where W_{hq} will have size $2h \times q$. Note that we could have different number of hidden units in each of the dimensions, which would alter these dimensions (i.e., it would not be $2h$ anymore)

Adding bidirectionality will largely increase computational costs, as gradients will have long dependency chains. We will explore this in the next lesson, where we will implement a bidirectional LSTM model!

Summary of Lesson 3

In Lesson 3, we have:

- ▶ Introduced formally Deep RNNs
- ▶ Discussed bidirectionality in the context of Hidden Markov models
- ▶ Explored the architecture and main computations of Bidirectional RNNs.

TO DO NEXT: This lesson does not have an associated Jupyter Notebook. Please advance to the next lesson.

In the next lesson, we will put into practice bidirectional LSTM networks by revisiting the BTC example from Lesson 2.